MILITARY SPECIFICATION

IGNITER ASSEMBLY FOR CARTRIDGES, 90MM, AP-T, M318A1 AND 90MM, TP-T, M353E1

1. SCOPE

1.1 This specification covers the loading, assembly, and packing of the igniter assembly for Cartridges, 90MM, AP–T, M318A1 and 90MM, TP–T, M353E1.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL–P–223 — Powder, Black

MIL–A–2550 — Ammunition and Special Weapons, General Specification for

MIL–I–45607 — Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes (ABC–STD–105)

MIL–STD–109 — Quality Assurance Terms and Definitions

MIL–STD–1235 — Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ORDNANCE CORPS

20–4–618 — Marking Diagram and Sealing of Steel Drums for Shipment of Propellants and Bag Charges

76–1–1039 — Drum, Metal, for 75MM, and 105MM Howitzer Propelling Charges

76–4–46 — Box, Packing with Metal Liner, M24, for Smokeless Powder, Assembly and Details

76–4–56 — Box, Packing, Metal-

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Wood, M17 for Smokeless Propellant

76–4–59 — Box, Packing, Metal-Wood, M18 for Smokeless Powder

7548683 — Igniter Assembly (Alternative)

7548695 — Igniter Assembly

7549033 — Container, Metal, Universal, M25 for Propellants and Explosives

7549036 — Marking Diagram and Sealing of Container, Metal, Universal M25 for Shipment of Propellant and Explosives

8858848 — Marking Diagram and Sealing of Metal Lined Wooden Packing Boxes for Shipment of Propellant

PUBLICATIONS

ORDNANCE CORPS

IEL–7548695 — Index of Inspection Equipment Lists

(*Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement function should be obtained from the procuring activity or as directed by the contracting officer.*)

3. REQUIREMENTS

3.1 Material. The materials shall be in accordance with applicable drawings and specifications.

3.2 Igniters. The igniters shall comply with all requirements specified on Drawing (dwg.) 7548683 or 7548695, and with all requirements specified in applicable specifications.

3.3 Moisture content of black powder. The moisture content of the black powder at the time the igniter assemblies are sealed in the packing container shall not exceed 0.60 percent when tested as specified in 4.3.3.

3.4 Workmanship. The igniters shall be free of dirt, grease or other foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL–STD–109 in order to define terms used herein. The provisions of Specification MIL–A–2550 shall apply.

4.1.1 *Submission of product.* At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

(a) A statement that the lot complies with al lof the quality assurance provisions specified within this specification.

(b) Number of units of product inspected.

(c) Results obtained, by defect code, for all inspection performed.

(d) Drawing, specification number and date, together with an identification and date of changes.

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(e) Certificates of compliance on all material purchased by the contractor when such material is controlled by Government specifications referenced in any of the contractual documents.

(f) Number of items in the lot.

(g) Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.2 *Government verification.* Using the contractor's written quality assurance procedure, this detail specification, the applicable drawings and other contractual documents as a guide, the Government inspector shall verify at unscheduled intervals all quality assurance operations performed by the contractor. Verification will be performed to the extent necessary to assure compliance with the contractual requirements. Severity of Government inspection of individual characteristics will be directly related to the seriousness of the classification assigned. In no instance will a characteristic classified "critical" be accepted solely on the basis of the contractor's records.

4.2 Inspection provisions.

4.2.1 *Lot formation.* A lot shall consist of igniter assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification, and same specification revision. Drawing, specification, and process changes not affecting safety, performance or fit as determined by the Government shall not necessitate changing the lot number. Product shall be submitted in accordance with MIL–STD–105 (or MIL–STD–1235, when applicable). Each lot shall contain:

(a) Black powder from not more than one lot.

(b) Igniter bags of not more than one type of material.

4.2.2 *Examination.* Inspection for Critical defects shall be 100 percent. Sampling plans and procedures for Major and Minor defects shall be in accordance with MIL–STD–105 except that continuous sampling plans in accordance with Standard MIL–STD–1235 may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to individual characteristics listed using an AQL of 0.65 percent for each Minor defect and an AQL of 0.40 percent for each Major defect. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 *Igniter assembly* (see dwg. 7548683 or dwg. 7548695):

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Stitches missing or broken to extent that power can escape	Visual	01001
102.	Charge weight out of tolerance	Scale	01002
Minor:	AQL 1.0 percent		
201.	Retaining tape missing	Visual	01003
202.	Evidence of poor workmanship (see 3.4)	Visual	01004

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4.2.2.2 *Box or container, prior to filling* (see dwgs. 76–4–46, 76–4–56, 7549033 and 76–1–1039 (as applicable)).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Crack or holes in metal liner (as applicable)	Visual	02001
102.	Foreign matter on interior	Visual	02002
Minor: None defined			

4.2.2.3 *Sealed box or container* (see dwgs. 76–4–46, 76–4–56, 76–4–59, 7549033, 8858848, 7549036, 20–4–618 and 76–1–1039 (as applicable)).

Categories	*Defects*	*Method of Inspection*	*Code No.*
Critical: None defined			
Major:	AQL 0.65 percent		
101.	Cover incompletely closed	Visual-Manual	03001
102.	Gasket missing or broken	Visual	03002
103.	Wooden cover split or otherwise damaged (when applicable)	Visual	03003
Minor:	AQL 2.5 percent		
201.	Car seal missing, unsealed or improperly positioned	Visual-Manual	03004
202.	Crack or hole in metal container	Visual	03005
203.	Hardware improperly engaged	Visual	03006
204.	Marking misleading or unidentifiable	Visual	03007
205.	Bare areas in exterior coating of metal containers the sum of which is in excess of ¼ square inch	Visual	03008

4.2.3 *Testing*

4.2.3.1 *Moisture content of black powder* (see 3.3) *Major defect, Code No. 01005.* The contractor shall provide adequate controls to assure the compliance of each lot with the applicable requirement. The contractor shall test black powder from one sample igniter from each eight hours of production. The sample shall be selected immediately prior to packing. The black powder shall be tested as specified in 4.3.2 and any failure to comply with the applicable requirements shall be cause for rejection of the eight hours production represented by the sample.

4.2.3.2 *Air pressure test* (see dwg. 8858848) *Major defect, Code No. 03009.* The containers shall be tested 100 percent in accordance with 4.3.1. If any container fails to meet the requirements it shall be rejected and removed from the lot.

4.2.4 *Inspection Equipment.* Index of Inspection Equipment Lists Number IEL–7548695 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in paragraph 6.3.

4.2.4.1 *Government rights to documentation.* Inspection equipment drawings and lists provided and revised in accordance with the requirements of the IEL may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspection equipment drawings are needed.

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4.2.4.2 *Supply and maintenance.* Supply and maintenance of the equipment listed on the IEL shall be in accordance with Specification MIL-I-45607.

4.2.4.3 *Government use of contractor's inspection and test equipment.* The contractor shall make available to the Government all inspection and test equipment necessary for determining conformance with contract requirements. Personnel for operating the equipment, and verification of its accuracy, shall be supplied by the contractor for the performance of examination or test by the Government.

4.3 Test methods and procedures.

4.3.1 *Air pressure test for containers* (see dwg. 8858848). The contained shall be subjected to an internal pressure of ½ to 1 pound per square inch. A water manometer shall be assembled in the system. A drop of 0.7 inch or more on the manometer in 15 seconds shall be cause for rejection.

4.3.2 *Moisture content of black powder.* Moisture content shall be determined in accordance with the procedure specified in Specification MIL-P-223.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.

5.1.1 *Level B.* The igniters shall be packed in containers complying with drawings 76-1-1039, 76-4-46, 76-4-56, 76-4-59 or 7549033.

5.2 **Marking.** The containers shall be marked in accordance with drawings 20-4-618, 7549036 or 8858848 (as applicable).

6. NOTES

6.1 Ordering data. Procurement documents shall specify the following:

(a) Title, number and date of this specification.

(b) Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection code numbers. The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment. The contractor shall design inspection equipment as required by the referenced Inspection Equipment Lists (IEL) in accordance with the instructions of paragraphs 6.3.1 through 6.3.7.

6.3.1 *Inspection equipment lists.* Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (OO Form 1242-3) the numbers of drawings of existing equipment designs or codes as indicated in paragraph 6.3.2. Design action required of the contractor with respect to the different types of drawings that may be listed is described in paragraphs 6.3.3 and 6.3.4. Action required by the contractor with respect to commercial inspection equipment is described in 6.3.5. The contractor will be required to prepare detailed drawings in accordance with 6.3.6 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.2 *Inspection equipment list codes.* The inspection equipment as defined in 6.3.3, 6.3.4, 6.3.5 and 6.3.6 will be designated in the IEL by the following codes:

CDAF — Contractor's design responsibilitn on Army format in accordance with MIL-D-45608.

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CDCF — Contractor's design responsibility on contractor format.

AD — Army design.

ADMU — Army design, mandatory for use.

CE — Commercial equipment.

SCD — Specification control drawing.

6.3.3 *Army designs.* Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications, mandatory and non-mandatory. When an inspection equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs and with the approval of the Government, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.7.

6.3.4 *Specification control drawing.* Specification control drawings depict the minimum equipment requirements in outline, descriptive, diagrammatic or pictorial form only and specify the required performance or other characteristics. Contractors must prepare detailed drawings (see 6.3.6) of their designs in support of specification control drawings. These contractor prepared designs must be approved by the Government prior to the fabrication or procuring of the equipment. Commercial equipment meeting the requirements of specification control drawings may be approved if described in sufficient detail to permit identification and evaluation by the Government. Designs shall be submitted for approval as specified in 6.3.7.

6.3.5 *Commercial equipment.* Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. Lists shall be submitted for approval to the inspection element of the agency administering the contract.

6.3.6 *Contractor designs.* Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Contractor designs identified as CDCF may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above. Contractor designs identified as CDAF shall comply with the format and requirements of MIL–D–45608, and in addition, contain the detail and information specified above. Final approved contractor designs identified as CDCF shall be suitable for microfilming. They must be non-reproducible diazo or electrostatic prints of top quality full size with black line on white background and shall provide a legible print produced via the 35MM microfilm process. Contractor shall submit one copy of the final design as a rolled set or flat set.

6.3.7 *Submission of contractor designs.*

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Designs shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND. Design review will normally be accomplished within one month after receipt by Picatinny Arsenal. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs.

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Preparing activity:

Army—MU

Project No. 1315-A457

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